META-BRAIN

From Prototype to Jarvis

The Complete Roadmap

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CONFIDENTIAL — INTERNAL ROADMAP

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OVERVIEW: WHAT WE ARE BUILDING

The Meta-Brain is a new kind of AI architecture. Not a better chatbot. A fundamentally different system that simulates before it speaks, remembers everything permanently, and over time becomes a personal Jarvis — an AI that knows your life, anticipates your needs, and reasons like a human.

■ The core insight: before answering anything, spawn 6 inner simulations simultaneously, let them vote on the best answer, and permanently store what was learned. This is not how any AI works today.

The Difference From Today's AI

Today's LLM	Meta-Brain
Single forward pass — predict next token	Six parallel simulations — then vote
Memory resets every session	Memory compounds forever — gets better with use
Can hallucinate freely	M5 catches hallucinations before output
Generic for everyone	M2 Digital Twin learns your specific life
No imagination	M4 generates scenarios that never existed
Reactive — only responds	Eventually: proactive — initiates from curiosity

The Six Inner Modules

Module	Name	Function	Key Equation
M1	World Model	Physical consequence simulation	p1 = exp(-D_KL)/Z
M2	Digital Twin	Bayesian model of your life	P(U|D_t) update
M3	Brain Simulation	Predicts your reaction	P(r=positive|o,state)
M4	Synthetic Reality	Counterfactual scenarios	S_cf = W0 + delta
M5	Sim Checker	Hallucination detector	T(o) threshold test
M6	Memory Box	Compounding universe memory	Q(t)=1-exp(-lt)

End State — What Jarvis Looks Like

Jarvis Capability	Meta-Brain Module That Delivers It
Knows your life completely	M2 Digital Twin tracks everything about you
Anticipates before you ask	M3 Brain Sim predicts what you need next

Never forgets anything	M6 Memory Box compounds forever
Never hallucinates	M5 gates every output before you hear it
Understands physical reality	M1 checks every claim against physics
Speaks, sees, acts	Voice + vision + tools + real-world actions
Runs your life proactively	Schedules, researches, executes autonomously

WHERE WE ARE NOW

■ PAPER WRITTEN ■ MATH PROVEN ■ NOTEBOOK RUNNING ■ PROTOTYPE BUILDING

What	Status
Research paper (17 pages)	Written — needs references completed
Mathematical framework	All theorems proven — 12 sections
Jupyter notebook	All 6 modules running — equations verified
Python prototype	M1 + M5 + M6 + Decision Engine — API based
Benchmark results	NOT YET — needed for arXiv credibility
Own trained model	NOT YET — Phase 5 goal
Voice / multimodal	NOT YET — Phase 7 goal
Full Jarvis	NOT YET — Phase 8 goal

■ HONEST ASSESSMENT: You have the blueprint, the math, and a small working prototype. The gap to full Jarvis is real but the path is clear. Every phase below closes one specific gap.

PHASE 1	Research Foundation	Paper + Math + Notebook	■ COMPLETE

Step 1: Complete the Research Paper

What: Finish references section and merge math document into original paper

How: Add all 10 references properly. Merge MetaBrain_Mathematical_Framework.docx into paper. Remove all "TO BE COMPLETED" placeholders.

Deliverable: Single complete PDF ready for arXiv submission

Step 2: Publish on arXiv

What: Submit paper to arXiv cs.AI category for public timestamp

How: Find one endorser in cs.AI (post on r/MachineLearning with your notebook link). Submit to https://arxiv.org. Category: cs.AI or cs.NE.

Deliverable: arXiv paper ID — public timestamp on your idea

Step 3: GitHub Repository

What: Make all code public to attract collaborators

How: Create github.com/PhenexAI/MetaBrain. Upload paper PDF, math notebook, prototype notebook. Write clear README.

Deliverable: Public GitHub repo with star count growing

PHASE MILESTONE	Paper live on arXiv + code on GitHub	COST	$0	TEAM	You alone

PHASE 2

API Prototype

M1 + M5 + M6 Working via API

2-4 WEEKS

Step 1: Fix API Connection

What: Get MetaBrain_v3.ipynb running end-to-end without errors

How: Get API key from console.anthropic.com. Paste in Cell 1. Run all cells. Confirm "API connected OK" in Cell 3.

Deliverable: All 10 cells running without errors

Step 2: Test on 50 Questions

What: Run Meta-Brain on 50 diverse questions and compare to baseline LLM

How: Use Cell 9 (comparison cell) for each question. Record: question, baseline answer, Meta-Brain answer, final score. Build a spreadsheet of results.

Deliverable: Spreadsheet with 50 comparison results

Step 3: Measure Accuracy Improvement

What: Quantify how much better Meta-Brain is vs single LLM pass

How: Have 3 people rate each answer pair blind (not knowing which is Meta-Brain). Calculate win rate. Should be close to Prediction P1: +22.96%.

Deliverable: Measured accuracy improvement number — your first real result

Step 4: Add M2 and M3 to Prototype

What: Extend prototype with Digital Twin and Brain Simulation modules

How: Add user preference learning to M2. Add reaction prediction to M3. Update Decision Engine weights to include all 5 modules.

Deliverable: Five-module prototype running

PHASE MILESTONE	Working 5-module prototype with measured results	**COST**	$10-50/month API costs	**TEAM**	You + 1 technical friend

PHASE 3

Benchmark Validation

Prove P1 on ARC-AGI

1-2 MONTHS

Step 1: Set Up ARC-AGI Benchmark

What: Run Meta-Brain on the ARC-AGI benchmark — the hardest test for novel reasoning

How: Download ARC-AGI dataset from github.com/fchollet/ARC-AGI. Write evaluation script that runs each task through Meta-Brain and baseline. Start with 100 tasks.

Deliverable: Evaluation script running on ARC-AGI

Step 2: Run Baseline Comparison

What: Compare Meta-Brain accuracy vs plain LLM on same ARC-AGI tasks

How: Run both systems on identical 100 ARC-AGI tasks. Record accuracy for each. Calculate improvement.

Deliverable: Numbers: Meta-Brain X% vs Baseline Y% on ARC-AGI

Step 3: Document Results

What: Write up results as an addendum to the paper

How: If Meta-Brain outperforms baseline even by 5%, write it up. This transforms the paper from theory to evidence. Post results on arXiv as v2 of paper.

Deliverable: Updated arXiv paper with real benchmark results

PHASE MILESTONE	Real benchmark numbers proving architecture works	**COST**	$50-200 compute	**TEAM**	You + 1 ML engineer

PHASE 4	**Full 6-Module Integration**	All Modules Connected	**2-4 MONTHS**

Step 1: Build M4 Synthetic Reality

What: Implement counterfactual scenario generation properly

How: Connect M4 to M1 world model. M4 perturbs initial conditions of M1 and generates N scenarios. Extract novel inferences per scenario.

Deliverable: M4 generating genuine counterfactual scenarios

Step 2: Build Full Universe Memory Box

What: Upgrade M6 from simple memory to full compounding system

How: Implement vector database (use ChromaDB — free, open source). Store compressed simulation outputs as vectors. Implement cosine similarity retrieval. Verify Q(t) improves over RAG.

Deliverable: M6 with vector storage outperforming RAG baseline

Step 3: Connect All 6 Modules

What: Build unified pipeline where all 6 modules run truly in parallel

How: Use Python asyncio for parallel execution. All 6 modules score simultaneously. Decision engine aggregates. Full o* = argmax SUM w_i p_i running live.

Deliverable: Full 6-module pipeline with parallel execution

Step 4: Performance Optimization

What: Make the system fast enough to use in real conversations

How: Target: under 10 seconds per response. Cache M1 world states. Batch M5 checks. Reduce API calls with smart routing.

Deliverable: System responding in under 10 seconds

PHASE MILESTONE	Complete 6-module Meta-Brain running	**COST**	$200-500/month	**TEAM**	You + 2 engineers

PHASE 5

Own Model Training

PhenexAI Base Model

6-12 MONTHS

Step 1: Decide on Model Strategy

What: Choose between training from scratch vs fine-tuning open source model

How: Option A: Fine-tune Llama 3 (70B) on simulation-specific data — faster, cheaper. Option B: Train own model from scratch — full control, years of work. Recommendation: Option A first.

Deliverable: Decision made + training plan written

Step 2: Build Training Data Pipeline

What: Create dataset of simulation-style reasoning examples

How: Generate 100,000 examples of: question + 6 module outputs + correct voting decision. Use current API prototype to generate these automatically.

Deliverable: Training dataset of 100K simulation examples

Step 3: Fine-tune Base Model

What: Train the model to natively perform Meta-Brain-style reasoning

How: Use Hugging Face + 8x A100 GPU cluster (rent from Lambda Labs ~$10/hr). Fine-tune for 1-2 weeks. Evaluate on ARC-AGI.

Deliverable: PhenexAI-1 model checkpoint

Step 4: Replace API with Own Model

What: Swap Claude API calls with own trained model

How: Update call_claude() to use local model. Compare performance to API baseline. This removes API dependency — full ownership.

Deliverable: Prototype running on own model — no API dependency

PHASE MILESTONE	PhenexAI owns its own trained model	**COST**	$50,000-200,000 compute	**TEAM**	You + 3 ML researchers

PHASE 6

Memory + Personalization

The Jarvis Layer

3-6 MONTHS

Step 1: Build Full Digital Twin

What: M2 that genuinely knows your life — work, projects, relationships, goals

How: Connect to calendar, email, notes (with permission). Build user model that learns from every interaction. Implement Bayesian update: $P(U|D_t)$ growing with every session.

Deliverable: M2 that knows your specific life context

Step 2: Persistent Universe Memory Box

What: Memory that survives across sessions and grows permanently

How: Store memory vectors in persistent database (PostgreSQL + pgvector). Every session adds to the box. Nothing ever forgotten. $Q(t)$ growing toward 1.0 over months of use.

Deliverable: Memory that persists and compounds across all sessions

Step 3: Proactive Intelligence

What: System that notices and acts without being asked

How: Build trigger system: if M3 predicts you need something, system surfaces it. "You have a meeting in 2 hours — here is the briefing you will need." "Based on your project, you should know about this paper published today."

Deliverable: System making proactive suggestions without prompting

Step 4: Personal Learning Loop

What: System that gets better the more you use it

How: Track which answers you accept vs reject. Update M2 weights based on your preferences. System accuracy improving measurably month over month.

Deliverable: Measurable improvement in personal relevance over time

PHASE MILESTONE	Jarvis knows your life and never forgets	**COST**	\$500/month infrastructure	**TEAM**	You + 1 backend + 1 ML

PHASE 7

Voice + Multimodal + World

Jarvis Speaks and Sees

3-6 MONTHS

Step 1: Voice Interface

What: Talk to Meta-Brain naturally — it talks back

How: Integrate Whisper (speech-to-text) + ElevenLabs (text-to-speech). Wake word detection. Natural conversation flow. Response latency under 3 seconds.

Deliverable: Voice conversation working end-to-end

Step 2: Vision — See the World

What: Meta-Brain can see images, screens, documents

How: Integrate vision model (GPT-4V or own fine-tuned). Screenshot analysis. Document reading. "Here is my screen — what should I do next?"

Deliverable: Vision-capable Meta-Brain

Step 3: Tool Use — Act in the World

What: Meta-Brain can take actions not just give answers

How: Build tool layer: web search, email sending, calendar management, file creation, code execution. Every action goes through M5 safety check first.

Deliverable: Meta-Brain executing real-world actions safely

Step 4: Mobile App

What: Jarvis in your pocket at all times

How: Build iOS + Android app. Always-on microphone option. Push notifications from proactive intelligence.

Deliverable: App live in App Store and Google Play

PHASE MILESTONE	Full multimodal Jarvis — voice, vision, action	**COST**	$2,000-5,000/ month	**TEAM**	You + 5-8 engineers

PHASE 8

Full Jarvis

Autonomous AGI Personal Assistant

12-24 MONTHS

Step 1: Autonomous Goal Pursuit

What: Give Jarvis a goal — it figures out and executes all the steps

How: Build goal decomposition: break high-level goal into subtasks. Execute each subtask using tool layer. Report back only when done or blocked.

Deliverable: Jarvis completing multi-step tasks autonomously

Step 2: World Model Grounded in Reality

What: M1 connected to real-time data — not just training knowledge

How: Connect M1 to live data: news, weather, markets, your calendar, your location. World model updates continuously. Predictions grounded in what is actually happening now.

Deliverable: Real-time grounded world model

Step 3: Multi-Agent Collaboration

What: Multiple Meta-Brain instances working together on complex problems

How: Spawn specialist agents: one for research, one for writing, one for planning. Agents share Universe Memory Box. Coordinate via message passing.

Deliverable: Multi-agent Meta-Brain swarm

Step 4: Full AGI Evaluation

What: Test on every major AGI benchmark

How: ARC-AGI, MMLU, HumanEval, BIG-Bench. Publish results. Compare to GPT-5, Gemini Ultra, Claude. This is the moment Meta-Brain proves itself.

Deliverable: Meta-Brain matching or exceeding frontier models on AGI benchmarks

PHASE MILESTONE	Full Jarvis — personal AGI assistant	**COST**	$10,000+/month	**TEAM**	Full team of 15-20

FUNDING STRATEGY

You do not need funding to reach Phase 3. Phases 1-3 cost under $500 total. Once you have benchmark results, the funding conversation changes completely.

Phase	Cost	Source	What You Build
Phase 1-2	$0-50	Self-funded	Paper + API prototype
Phase 3	$50-200	Self-funded	Benchmark results
Phase 4	$500-2K	Grants / Angels	Full 6 modules
Phase 5	$50K-200K	Seed round	Own model
Phase 6-7	$500K-2M	Series A	Jarvis layer + multimodal
Phase 8	$5M+	Series B+	Full AGI assistant

Canadian Funding Sources — Apply Now

Funding Source	Amount + When to Apply
NSERC Alliance Grant	Up to $500K — Canadian AI research. Apply after arXiv paper published.
Next AI (Montreal)	Equity-free $100K + mentorship. Apply with paper + prototype.
Creative Destruction Lab	Equity + network. Strong AI focus. Apply after Phase 3.
NRC IRAP	Up to $50K non-dilutive for early tech companies. Apply immediately.
BDC Seed Fund	Up to $500K seed. Apply after Phase 4 demo.
Mitacs Accelerate	Subsidized researcher salary. Partner with UBC or SFU.

TEAM PLAN — WHO YOU NEED

Phase	Role	What They Do	Cost
Phase 1-3	You	Founder + researcher	Free
Phase 2-3	ML Engineer (part-time)	Prototype + benchmarks	Equity or $2-5K/month
Phase 4-5	Senior ML Researcher	Model training lead	$8-12K/month or equity
Phase 4-5	Backend Engineer	Infrastructure + APIs	$6-10K/month
Phase 6-7	Mobile Developer	iOS + Android app	$6-8K/month
Phase 6-7	Product Designer	UX for Jarvis interface	$5-8K/month
Phase 7-8	Voice/Audio Engineer	Speech pipeline	$7-10K/month
Phase 8	Head of Research	Lead AGI research team	$15-20K/month

■ WHERE TO FIND TECHNICAL CO-FOUNDERS: Post your GitHub repo on r/MachineLearning. Message PhD students at UBC/SFU working on world models or cognitive architectures. Post on LinkedIn with your arXiv paper. The paper is your recruiting tool.

COMPLETE TECH STACK

Component	Technology + Notes
Base LLM (now)	Claude API (Anthropic) — anthropic Python library
Base LLM (Phase 5)	Own fine-tuned Llama 3 70B — Hugging Face Transformers
Vector Memory (M6)	ChromaDB (Phase 4) → pgvector + PostgreSQL (Phase 6)
World Model (M1)	Physics simulation: custom + NVIDIA Cosmos API
Speech-to-text	OpenAI Whisper (open source, free)
Text-to-speech	ElevenLabs API (Phase 7)
Vision	GPT-4V or own fine-tuned vision model
Parallel execution	Python asyncio + concurrent.futures
Backend API	FastAPI (Python) — fast, modern, well-documented
Database	PostgreSQL + pgvector for memory storage
Mobile app	React Native (one codebase for iOS + Android)
Model training	Hugging Face + Lambda Labs GPU rental
Experiment tracking	Weights & Biases (free for small teams)
Deployment	Docker + AWS or Google Cloud
Notebooks	Jupyter + Google Colab (free GPU)
Version control	GitHub — github.com/PhenexAI

RISK MAP — WHAT COULD GO WRONG

Risk	Likelihood	Mitigation	When
API costs too high	Medium	Switch to Ollama local model	Phase 2
Benchmark results disappointing	Medium	Refine module weights, try different LLM base	Phase 3
Someone publishes same idea first	Low	arXiv timestamp protects priority — publish NOW	Phase 1
Cannot find technical co-founder	Medium	GitHub + arXiv paper is best recruiting tool	Phase 2-3
Model training too expensive	High	Fine-tune open source instead of train from scratch	Phase 5
API provider changes pricing	Low	Own model in Phase 5 removes this dependency	Phase 5
Competition from big labs	High	Move fast — architecture IP + benchmark results matter	All
Running out of money	Medium	Phases 1-3 cost under $500 — reach Phase 3 first	Phase 1-3

MASTER CHECKLIST — NOTHING LEFT BEHIND

PHASE 1 — This Week

■ HIGH	■ Complete paper references section
■ HIGH	■ Merge math document into paper PDF
■ HIGH	■ Get Anthropic API key from console.anthropic.com
■ HIGH	■ Run MetaBrain_v3.ipynb end-to-end successfully
■ HIGH	■ Create GitHub repo: github.com/PhenexAI/MetaBrain
■ HIGH	■ Upload paper + notebooks to GitHub
■ MEDIUM	■ Post on r/MachineLearning asking for arXiv endorser
■ MEDIUM	■ Apply for NSERC or NRC IRAP grant

PHASE 2 — Next Month

■ HIGH	■ Run Meta-Brain on 50 test questions
■ HIGH	■ Build comparison spreadsheet: baseline vs Meta-Brain
■ HIGH	■ Add M2 Digital Twin to prototype
■ MEDIUM	■ Add M3 Brain Simulation to prototype
■ HIGH	■ Get arXiv endorser and submit paper
■ MEDIUM	■ Write README for GitHub with demo GIF
■ MEDIUM	■ Apply to Next AI or Creative Destruction Lab

PHASE 3 — 1-2 Months

■ HIGH	■ Download ARC-AGI benchmark dataset
■ HIGH	■ Run Meta-Brain on 100 ARC-AGI tasks
■ HIGH	■ Measure accuracy vs baseline — record the number
■ HIGH	■ Write up results — update arXiv paper to v2
■ MEDIUM	■ Post results on Twitter/X and LinkedIn
■ MEDIUM	■ Reach out to 5 ML researchers about collaborating

ONGOING — Every Month

■ HIGH	■ Update GitHub with latest code
■ MEDIUM	■ Track memory quality Q(t) growth

■ MEDIUM	■ Test new questions through Meta-Brain
■ HIGH	■ Document what works and what does not
■ HIGH	■ Keep applying for grants and accelerators

THE SINGLE MOST IMPORTANT THING

Get the API key. Run the notebook. Post on GitHub. Then post on r/MachineLearning asking for an arXiv endorser. Everything else flows from those four actions. The gap between where you are and full Jarvis is real. But every gap in this document has a specific step that closes it. You have the idea. You have the math. You have the code. Now build it.

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